Dynamic Resampling-Guided Multidirectional FreqFormer for Imbalanced Partially Labeled Hyperspectral Images

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***Abstract*—Hyperspectral image (HSI) classification plays a critical role in numerous remote sensing applications, such as precision agriculture, urban land-cover mapping, and ecological monitoring. However, real-world hyperspectral data usually exhibit both label ambiguity and class imbalance, posing a challenge to the algorithmic robustness of hyperspectral classification under partially labeled settings. Existing methods mainly focus on either partial label learning (PLL) for natural images or imbalance learning among the overall dataset independently while largely overlooking the HSI characteristics and intra-batch distribution variations among classes. Thus, we propose a dynamic resampling (DR)-guided multidirectional FreqFormer (MDFF) method under a coherent framework to overcome data imbalance and label ambiguity in the HSI simultaneously. First, we propose a DR strategy with class prototypes to alleviate the problem of class imbalance. Second, to address the limitations of traditional airspace attention mechanisms, we develop an MDFF based on fast Fourier transformation (FFT) to mine sufficient spectral information along the different directions of the HSI cube in the frequency domain, which can capture more valuable information. Finally, we introduce a smoothing PLL loss and a mutual-entropy loss for more precise and reliable label disambiguation. The proposed framework achieves the best classification performance when the partial label rate is set to 0.1, obtaining 96.30%, 94.91%, and 97.99% overall accuracy (OA) on Indian Pines, Houston13, and WHU-Hi-Honghu, respectively, demonstrating the superiority of the proposed methods compared with the existing method. The codes are available at https://github.com/Nemo96yu/DrPLL**

***Index Terms*—Class imbalance, hyperspectral image (HSI), partial labels, resampling.**

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I. Introduction

With the advancement of remote sensing technology, hyperspectral imaging technology can acquire spatial information and continuous spectral features of target objects simultaneously [1]. Hyperspectral images (HSIs) have the unique advantage of spatial–spectral integration to finely characterize target objects, demonstrating unparalleled technical value across defense reconnaissance, ecological surveillance, precision farming, and allied scientific disciplines [2], [3], [4]. HSI classification refers to identifying the land-cover type of the pixels in HSI, playing a vital role in the above tasks of HSI intelligent interpretation [5], [6]. However, due to the presence of pixel-confusing zones and uneven land-cover distribution, label ambiguity and class imbalance frequently pose significant challenges to HSI classification. In particular, label ambiguity manifests as spectral interclass undemarcation within pixel-confusing zones, compromising accurate ground truth determination through human–computer collaborative annotation [7], [8]. Concurrently, class imbalance stems from significant interclass sample quantity disparities in land-cover distribution, ultimately resulting in insufficient feature representation learning for minority classes during model optimization [9], [10]. To address these issues, researchers have proposed partial label learning (PLL) for label ambiguity and class-balancing strategies for class imbalance.

PLL originates from ambiguous annotation/partial labeled paradigms where each sample's label set contains the ground truth along with confusing candidate labels [11], [12]. Recent studies have witnessed the development of PLL frameworks that simultaneously reduce annotation costs and mitigate labeling complexity through disambiguation mechanisms. An effective PLL framework with semantic label representation (ParSE) is proposed to capture sufficiently distinguishable representations for promoting label disambiguation [13]. PLL with contrastive label disambiguation (PiCO) identifies the ground-truth label from the candidate set by using contrastively learned prototypes [14], [15]. To simplify and improve the contrastive process of PiCO, PLL with a guided prototypical classifier (PaPi) is proposed to contrast the prototypical similarity with the visual similarity between categories [16]. An asymmetric dual-task cotraining (AsyCo) PLL model learns from different views explicitly

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by training a disambiguation network and an auxiliary network via optimizing different tasks [17]. To combine PLL with HSI classification, a superpixelwise low-rank approximation (LRA)-based PLL method (SLAP) has been proposed [7]. To further improve the performance on partially labeled HSI classification, a cross-contrastive disambiguation (CrCD) method [18] and a max-gap loss-guided siamese reconstruction network (MAGS) [19] have been proposed to achieve high-quality feature extraction under coarse supervision. Despite their effectiveness, these methods generally assume a relatively balanced category distribution and therefore neglect the severe class imbalance commonly encountered in real-world HSI datasets. Consequently, the disambiguation process can be biased toward majority classes, resulting in degraded performance for minority categories.

Class-balancing strategies are proposed to systematically alleviate interclass imbalanced distribution in classification tasks, thereby preventing model bias toward predominant classes while preserving decision-boundary integrity [20], [21], [22]. As a simple but effective class-balancing method, sample resampling can balance the data distribution by undersampling samples from the majority class or oversampling samples from the minority class, thus improving the classification performance to a certain extent [23], [24], [25], [26]. However, oversampling paradoxically induces overreplication of minority-class samples during training iterations while undersampling risks critical information erosion in majority-class distributions, both scenarios potentially compromising model robustness through distorted decision boundaries [27], [28], [29]. Recent studies have seen a growing trend of reweighting the loss function of the classifier according to class distribution ratios within the dataset, which pays more attention to the classification of minority classes and achieves undistorted decision boundaries [30], [31], [32]. Nevertheless, the weighted loss strategy requires accurately labeled data for precise statistics on category distribution, which does not accommodate the class imbalance problem in partially labeled data. Although class-balancing strategies effectively alleviate classifier bias in conventional supervised learning, they remain inadequate for partially labeled HSI classification. In particular, most existing methods rely on accurate class statistics derived from clean labels, which are unavailable under label ambiguity.

From the above discussion, the critical research gap is that current studies typically treat label ambiguity and class imbalance as independent problems, whereas these two challenges are intrinsically coupled in partially labeled HSI classification. Label ambiguity hinders reliable estimation of class distributions, while class imbalance further aggravates the label disambiguation process by biasing model optimization toward dominant classes. Recent advances in natural image classification have shifted attention to tackling class imbalance under partially labeled supervision, with novel frameworks emerging to handle ambiguous annotations and imbalanced data distributions jointly. The main challenge of partially labeled imbalanced classification is how to accurately estimate the class distribution of data with label ambiguity. To overcome this challenge, a Sinkhorn label refinery (SoLar) framework derives an optimal transport objective that refines the pseudolabels to match the true class prior [33]. A dynamic rebalancing method introduces a parametric class distribution without requiring the prior of the oracle class distribution [34]. To balance the performance of the majority class and minority class, a model in [35] trains two classifiers that excel in different class shots and combines their predictions automatically. These methods were originally designed for natural images and only rely on ResNet to extract spatial features, so they overlook the information in the continuous spectral dimension of hyperspectral data and fail to capture spectral–spatial features of HSI. Thus, they cannot achieve satisfactory results when directly applied to the task of HSI classification.

As mentioned above, imbalanced classification of partially labeled HSI poses a dual challenge: label ambiguity misguides class distribution statistics and the implementation of class balancing strategies, and imbalanced class distributions exacerbate errors in label disambiguation, leading to unsatisfactory classification performance. Currently, there is no coherent framework for imbalanced classification of partially labeled HSI which can adapt to the spatial–spectral structure of HSIs. Integrating existing HSI encoders [36], [37], [38] into the aforementioned framework offers a streamlined solution to mitigate label ambiguity and class imbalance inherent in HSI simultaneously. However, most of the existing HSI encoders only perform feature extraction in the traditional spatial domain, which cannot adequately capture discriminative features conducive to classification in the face of label ambiguity and class imbalance. Recently, some researchers have proposed to utilize the Fourier transform technique (FFT) to mine more valuable semantic features [39], [40], [41]. Nevertheless, most existing approaches perform Fourier transformation along a single spatial or spectral direction. Such a design only captures partial frequency responses and inevitably overlooks complementary information distributed across different orientations of the HSI cube. Given that hyperspectral data contain highly coupled spectral–spatial structures, discriminative patterns may manifest differently along horizontal, vertical, and spectral dimensions. Therefore, relying on a single-direction frequency representation often results in incomplete feature extraction and limited characterization of global dependencies. This limitation motivates the development of a multidirectional frequency learning mechanism capable of comprehensively exploring spectral–spatial information from multiple perspectives. Moreover, existing frameworks rectify class imbalance solely based on global class distribution of the entire training dataset while neglecting the impact of intrabatch class distribution variations on model training. In particular, random minibatch sampling causes significant differences in class distributions between batches, leading to inconsistent numbers of samples for each class within a batch. This distribution drift can introduce biased gradient estimates, affecting the stability of model training. Therefore, there's an urgent need for a resampling mechanism that can dynamically detect the class distribution of the current batch and adaptively adjust the sampling weights. Overall, it's imperative to design an effective and coherent framework with

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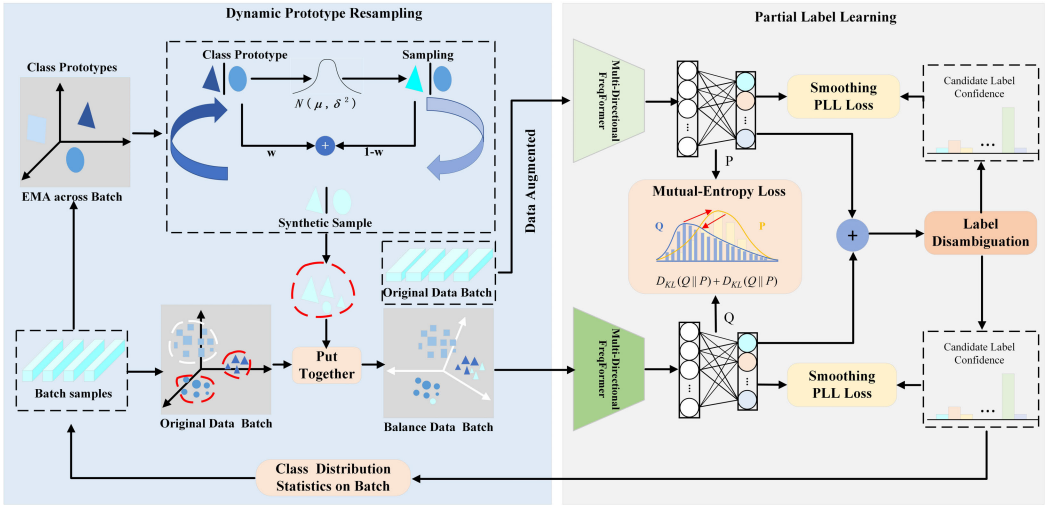


Fig. 1. Overall network architectures.

a high-quality feature encoder tailored to HSI characteristics for partially labeled HSI imbalanced classification.

In this article, we propose a **d**ynamic **r**esamping **p**artial **l**abel **l**earning (DrPLL) framework to solve the class imbalance and label ambiguity for the HSI classification task. A dynamic resampling (DR) multidirectional FreqFormer (MDFF) method is designed under the DrPLL framework for the first time. First, we propose a DR strategy to solve the challenge of class imbalance. In particular, we initialize updatable class prototypes through coarse computation, then dynamically monitor batch-wise distributions via label confidence metrics. For classes with below-average sample counts, it synthesizes virtual samples through Gaussian sampling centered at current prototypes. Both label confidence values and class prototypes undergo coordinated updates throughout iterative training. Second, inspired by the frequency domain-based self-attention solver (FSAS) [39], we develop the MDFF to mine sufficient spatial–spectral information along the different directions of the HSI cube. Finally, we introduce a smoothing PLL loss and a mutual-entropy loss, which alleviate the overfitting problem and steer the model toward more precise label predictions for label disambiguation. Overall, the main contributions of this article can be summarized as follows.

1) To address the class imbalance in HSI PLL, we devise a DR strategy. This approach leverages Gaussian sampling around class prototypes to generate synthetic samples while dynamically refining these prototypes during training.
2) To improve the discriminative and efficiency of feature extraction, we introduce the MDFF to establish robust feature representations that form the foundation for subsequent label disambiguation and class distribution balancing.
3) For precise training in PLL, we design a smoothing PLL loss and a mutual-entropy loss specifically tailored to the unique requirements of partial label scenarios. The smoothing PLL loss is used to solve model overfitting, and the mutual-entropy loss prompts reliable label disambiguation for accurate class distribution estimation.
4) By combining the DR and MDFF, we propose a coherent framework named DrPLL to simultaneously address the label ambiguity and class imbalance in HSI classification. Extensive experimental results on benchmark datasets demonstrate the superior performance of the proposed methods on the partially labeled HSI classification task with class imbalance.

II. Proposed Model

The proposed DrPLL framework is composed of two parts: the dynamic prototype resampling part for class imbalance and the PLL part for label ambiguity. In particular, as shown in Fig. 1, in the dynamic sampling process, we sample data from a Gaussian distribution of category prototypes so that the number of samples from a minority category reaches the mean value in each training batch. In the PLL module, we also introduce the MDFF to provide high-quality features for label disambiguation and optimize the model parameters via smoothing PLL loss and mutual-entropy loss.

A. Problem Setting

Let $\mathcal{X}$ be the input space and $\mathcal{Y} = \{1, 2, \ldots, c\}$ be the class space. $x_i \in \mathbb{R}^{s\times s\times b}$ denotes the ith sample: an $s\times s\times b$ data cube cropped from the PCA-reduced HSI with b channels and patch size $s \times s$, centered at the ith pixel. The candidate label set space $\mathcal{S}$ is the nonempty power set of $\mathcal{Y}$, which is generated via flipping each incorrect label with a probability R called the partial label rate [16]. The higher the value of R is, the higher the label ambiguity is. An imbalanced partially labeled training HSI set can be denoted as $D_{train} = \{(x_i, S_i)\}_{i=1}^{N}$, where any x_i is associated with a candidate label set S_i including its invisible ground truth label and N is the total number of samples. The training set exhibits class imbalance, where the majority classes contain significantly more samples than the minority classes. The goal of imbalanced HSI PLL is to learn a deep network on the partially labeled and imbalanced training set D_{train}, thereby reducing annotation costs and mitigating labeling pressures.

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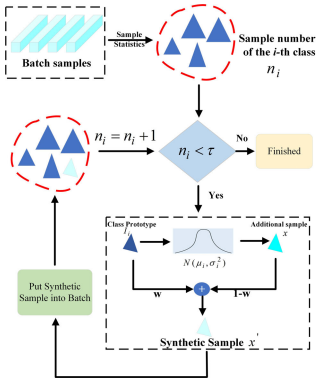


Fig. 2. Flowchart of DR.

Note that for PLL, we suppose $Q = \{q_1, q_2, \ldots, q_n\} \in \mathbb{R}^{N\times c}$ is the labeling confidence matrix, where $q_i = [q_{i1}, \ldots, q_{ic}] \in \mathbb{R}^c$ represents the labeling confidence vector of x_i, and q_{ij} denotes the probability of the jth label being the groun truth of x_i. As in the previous works [16], Q is initialized according to the candidate label set as follows:

$$q_{ij} = \begin{cases} \dfrac{1}{|S_i|}, & if\ j \in S_i \\ 0, & otherwise \end{cases} \tag{1}$$

where S_i denotes the candidate label set of ith sample. It should be noted that Q would be updated according to the classification results during the training process, and only the confidence of the candidate label has a chance to be positive. For the ideal state, the confidence of the hidden ground truth will be close to 1, and others will be close to 0.

B. Dynamic Resampling

Given the training dataset D_{train}, by computing the average representation of the corresponding sample of each class, we maintain the class prototypes $L = \{I_1, I_2, \ldots, I_c\}$ as the class representative samples. The initial category prototypes can be obtained as follows:

$$I_i = \frac{1}{m} \sum_{i\in S_k} x_k \tag{2}$$

where I_i is the prototype of the ith class in the whole label space $\mathcal{Y}$, m is the total sample number where the candidate label set includes the ith class, and S_k is the candidate label set of the kth sample. Owing to the existence of label ambiguity, the ground truth of x_i is hidden in the candidate label set, making it difficult to obtain perfect class prototypes in the initial conditions. Thus, during the training process, L should be updated according to the labeling confidence matrix Q as follows:

$$I_i = \alpha I_i + (1-\alpha)\frac{1}{m}\sum_{k}^{m} q_{ki} x_k \tag{3}$$

where the pseudolabel of sample x_k is the label i (the max value in q_k is q_{ki}), m is the total number of samples with pseudolabel i. In this way, the class prototypes can be more and more representative and promote the further resampling process.

For each training batch $B = \{x_i, S_i\}_{i=1}^n$ loaded from D_{train}, we statistic the sample number of each class $\{n_1, n_2, \ldots, n_c\}$ and set the sampling threshold τ as n/c to determine which categories need additional samples. In particular, as shown in Fig. 2, if the sample number n_i of the ith class is lower than τ, we treat it as a minority class that needs additional samples to promote its feature extraction, and vice versa. Taking the minority class i as an example, we assume all samples in that category follow a Gaussian distribution $N(\mu_i, \delta_i^2)$, which is computed from its class prototype I_i. Based on this assumption, we sample the additional sample x' from the corresponding Gaussian distribution as follows:

$$x' \sim N\left(\mu_i, \delta_i^2\right). \tag{4}$$

Then, to ensure the representativeness of the additional sample, we compute the weighted sum of it and the I_i as a synthetic sample as follows:

$$x' = wI_i + (1-w)\,x' \tag{5}$$

where w is the weight value between 0 and 1. In other words, synthetic samples specifically refer to sample data synthesized by combining samples drawn from a Gaussian distribution with their corresponding class prototypes according to a specific ratio. Finally, we can obtain the synthetic sample set $\{(x'_k, y'_k)|y_k = i\}_{k=1}^{\tau-n_i}$ for the ith class via repeating the sampling in (4) and synthesizing in (5) $\tau - n_i$ times. In this way, we can put the synthetic sample set together with the original sample batch, enabling the sample number of the ith class to reach the threshold number of samples and achieving the class balancing effect. Conversely, for the majority class data exceeding or equal to the threshold τ in each data batch, no additional sampling process is performed, thereby preserving its original data volume.

C. Multidirectional FreqFormer

Existing transformer-based HSI encoders usually compute the attention map in the traditional spatial domain, which is sensitive to pixel-level noise and results in the loss of valuable high-frequency detail information [39]. Therefore, a FSAS transforms the feature representation into the frequency domain via the Fourier transform technique (FFT) and efficiently estimates the attention map by an element-wise product operation. In particular, after obtaining the query Q, key K, and value V, it leverages the fast Fourier transform (FFT) to estimate the correlation of Q and K in the frequency domain by

$$A = \mathcal{F}^{-1}\left(\mathcal{F}(Q)\,\overline{\mathcal{F}(K)}\right) \tag{6}$$

where $\mathcal{F}(\cdot)$ denotes the FFT, $\mathcal{F}^{-1}(\cdot)$ denotes the inverse FFT, $\overline{\mathcal{F}(\cdot)}$ denotes the conjugated operation, A is the attention score. Then, it estimates the aggregated features by

$$V_{att} = LN(A)\,V \tag{7}$$

where a layer norm function $LN(\cdot)$ is used to normalize the attention socre A and V_{att} is the attention value. The final output features of FSAS are computed as follows:

$$X_{att} = X + f_{conv}^{1\times 1}(V_{att}) \tag{8}$$

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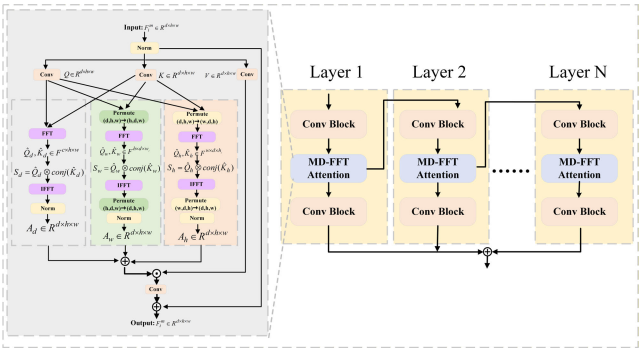


Fig. 3. Architecture of MDFF.

where $f_{conv}^{1\times1}(\cdot)$ denotes the convolution function with the kernel size of 1×1 and X_{att} is the output features of FSAS.

Although FSAS presents an effective and efficient method to explore the properties of Transformers for high-quality feature extraction, it focuses on natural images and ignores the spatial–spectral information hidden in HSI.

To this end, we develop an effective MDFF to extract high-quality features for partially labeled HSI imbalanced classification. In particular, the MDFF is composed of three layers of feature extraction blocks, where each layer contains a convolutional block and the multidirectional frequency domain attention block, thus fully and efficiently mining the spatial–spectral information hidden in HSIs. For each layer, as shown in Fig. 3, we take the outputs of the previous layer as its input and first put them into the convolution block for coarse spatial feature extraction as follows:

$$F_i^m = \mathcal{G}\left(\mathcal{D}\left(BN\left(f_{conv}^{3\times3}\left(F_i^{m-1}\right)\right)\right)\right) \tag{9}$$

where $\mathcal{D}(\cdot)$, $BN(\cdot)$, and $\mathcal{G}(\cdot)$ denote the dropout function, batch normalization function, and GELU activation function, respectively. F_i^{m-1} is the output of the $(m-1)$th layer for ith sample, F_i^m is the coarse feature representation of the ith sample computed from the convolutional block of mth layer. It is noted that the input of the first layer for the ith sample is the data cube x_i entered at the ith pixel. Then, we put the coarse feature representation F_i^m into the multidirectional frequency attention block for further spatial–spectral information extraction. As shown in Fig. 3, we obtain query Q, key K, and value V needed for attention calculation by a 1×1 point-wise convolution and 5×5 depth-wise convolution. Then, we apply the FFT along the channel direction, vertical direction, and horizontal direction to estimate the spectral correlation and spatial correlation between F_q^m and F_k^m in the frequency domain

$$\begin{cases} A_d = \mathcal{F}_d^{-1}\left(\mathcal{F}_d(Q)\,\overline{\mathcal{F}_d(K)}\right) \\ A_h = \mathcal{F}_h^{-1}\left(\mathcal{F}_h(Q)\,\overline{\mathcal{F}_h(K)}\right) \\ A_w = \mathcal{F}_w^{-1}(\mathcal{F}_w(Q)\,\overline{\mathcal{F}_w(K)} \end{cases} \tag{10}$$

where $\mathcal{F}_d(\cdot)$ denotes FFT function along the channel direction, $\mathcal{F}_h(\cdot)$ denotes FFT function along the vertical direction, and $\mathcal{F}_w(\cdot)$ denotes FFT function along the horizontal direction. $\mathcal{F}_d^{-1}(\cdot)$, $\mathcal{F}_h^{-1}(\cdot)$, and $\mathcal{F}_w^{-1}(\cdot)$ denote the inverse FFT along the channel direction, vertical direction, and horizontal direction, respectively. A_d, A_h, and A_w are the attention scores along the channel direction, vertical direction, and horizontal direction, respectively. After that, we obtain the spatial–spectral correlation features by

$$V_{att}^m = \left(LN\left(A_d\right) + LN\left(A_w\right) + LN\left(A_h\right)\right)V \tag{11}$$

where V_{att}^m is the overall feature that covers rich spatial and spectral characteristics.

To prevent the gradient from vanishing, we add V_{att}^m to the input features F_i^m to generate the outputs as follows:

$$F_i^m = F_i^m + f_{conv}^{1\times1}\left(V_{att}^m\right). \tag{12}$$

Finally, we fuse the outputs of each layer to obtain the comprehensive and discriminative feature representation F_i for each sample by

$$F_i = \sum_{m=0}^{M} \mathcal{G}\left(MP\left(BN\left(f_{conv}^{5\times5}\left(F_i^m\right)\right)\right)\right) \tag{13}$$

where $MP(\cdot)$ denotes the max pooling process. In this way, we can capture advanced semantic representation as the number of network layers increases while ensuring that the original semantics are not lost, obtaining more robust feature representation. In the same way, in each batch, we put the synthetic samples into the MDFF together with the original sample for class balancing and obtain the corresponding feature representations. For the ith synthetic sample x_i' in each batch, the feature representation is denoted as V_i.

To learn more discriminative features, for each sample x_i, we generate the augmented data cube x_i^a through the Gaussian bluring as follows:

$$x_i^a(x,y) = x_i G_\sigma(x,y) \tag{14}$$

where $x_i^a(x,y)$ represents the feature value at spatial position (x,y) of the ith sample's feature map after Gaussian blur. G_σ is the Gaussian blurring function defined as follows:

$$G_\sigma(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \tag{15}$$

where (x,y) denotes the pixel coordinate offset in the patch plane. σ is the standard deviation of the Gaussian distribution, which controls the level of blur. After that, we obtain the augmented feature representation F_i' of the augmented data cube x_i^a through a weight-shared encoding process of (13). In this way, we can further enhance the representation learning process.

D. Label Disambiguation

Given the learned feature F_i from ith original sample x_i and F_i' from the augmented data of x_i^a, we leverage the weight-shared linear classifiers for label prediction and obtain the label probabilities $p_i = \{p_{i1}, p_{i2}, \ldots, p_{ic}\}$ and $p_i' = \{p_{i1}', p_{i2}', \ldots, p_{ic}'\}$ as follows:

$$\begin{cases} p_i = LN\left(F_i\right)W \\ p_i' = LN\left(F_i'\right)W \end{cases} \tag{16}$$

where W is the learnable parameter matrix, p_{ij} denotes the predicted label probability of the ith sample for the jth label, and p_{ij}' denotes the predicted label probability of the ith augmented sample for the jth label. As mentioned in the

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subsection on problem setting, we need to conduct label disambiguation for PLL and capture the information from label ambiguity for modal updating. Thus, we leverage the sum of p_i puls p_i' to achieve the label disambiguation as follows:

$$q_{ij} = \begin{cases} \dfrac{exp\left(p_{ij} + p'_{ij}\right)}{\sum_{k \in S_i} exp\left(p_{ik} + p'_{ik}\right)}, & if\ j \in S_i \\ 0, & otherwise \end{cases} \tag{17}$$

where $exp(\cdot)$ is the exponential function and q_{ij} represents the labeling confidence of jth label of ith sample which is initialized in (1). In this way, the confidence value of the ground-truth label would become higher and progressively closer to 1, and vice versa.

E. Loss Function

To mitigate the overfitting phenomenon in PLL, we design a partial label smoothing loss, whose basic idea is to smooth the probability distribution of the label confidence matrix, thus improving its generalization ability. In partifcular, we transform the original label confidence q_i into the following smoothed labels:

$$\overline{q_i} = (1 - \theta)\, q_i + \theta q_i' \tag{18}$$

where q_i' is the initial label confidence of the ith sample computed via (1), θ is the smoothing parameter, and $\overline{q_i}$ is the smoothed label confidence vector of ith sample. According to the definition of cross entropy, the formula for partial label smoothing loss can be obtained

$$L_{cls} = -\sum_{j=1}^{c} \overline{q_{ij}} \left(\log \left(\frac{exp\left(p_{ij}\right)}{\sum exp\left(p_{ij}\right)} \right) + \log \left(\frac{exp\left(p'_{ij}\right)}{\sum exp\left(p'_{ij}\right)} \right) \right) \tag{19}$$

where $exp(\cdot)$ represents the natural exponential function and L_{cls} is the partial label smoothing loss.

Furthermore, to further establish the connection between weight-sharing classifiers and obtain more precise classification results, we design the mutual-entropy loss based on information entropy as follows:

$$L_{mue} = \frac{1}{2}(\sigma\left(p_i\right) \log\left(\sigma\left(p_i'\right)\right) + \sigma\left(p_i'\right) \log\left(\sigma\left(p_i\right)\right) \tag{20}$$

where $\sigma(\cdot)$ is the softmax function and L_{mue} is the mutual-entropy loss. In this way, we establish consistency constraints between the weight-shared classifiers so that they can provide a more reliable basis for subsequent label disambiguation. For the feature representation V_i of the synthetic sample, the classifier outputs its corresponding predictive result p_i^s. Thus, we directly compute the cross-entropy loss as follows:

$$L'_{cls} = -\frac{1}{2} \sum_{i=1}^{c} y_i' \log\left(p_i^s\right) \tag{21}$$

where y_i' is the one-hot label of the synthetic sample, and L'_{cls} is the cross-entropy loss of the synthetic sample. Finally, the overall loss of the model can be computed by

$$Loss = L_{cls} + L_{mue} + L'_{cls} \tag{22}$$

where $Loss$ is the total loss.

III. Experimental Results and Discussion

In this section, we will verify the proposed method on three hyperspectral datasets and compare it with some of the state-of-the-art partially labeled imbalanced classification frameworks via combining several HSI encoders.

A. Datasets

1) The Indian Pines dataset is an HSI of a region with Indian Pines trees in Indiana, USA, which has a size of 145×145 pixels with 200 spectral bands ranging from 0.4 to 0.25 μm.
2) The WHU-Hi-Honghu dataset is an HSI of a region in Honghu City, Hebei Province, China, which has a size of 940×475 pixels with 270 spectral bands ranging from 400 to 100 nm.
3) The Houston 2013 dataset is an HSI of the University of Houston with its surrounding urban areas. The size of this image is 349×1905, which has 144 spectral bands and includes 15 types of land cover.

The above three datasets would be processed via PCA dimensional reduction and patch extraction, and the patch size of each sample is fixed as $13 \times 13 \times 30$. In the experiment, we randomly select some samples for each class as the training set, and the rest as the testing set, which is shown in Table I.

B. Experimental Setup

1) *Evaluation Indicators:* To quantitatively evaluate and analyze the superiority of the proposed method, four metrics are employed: overall accuracy (OA), average accuracy (AA), kappa coefficient (KC), and per-class classification accuracy. Higher values for these metrics indicate superior classification results.
2) *Comparative Approaches:* In the experiment, several state-of-the-art imbalanced classification frameworks for partially labeled data (e.g., Recordes [34], HTC-LTPLL [35]) with several transformer-based HSI encoders (e.g., SSFTT [36], MSSTT [38]) were selected for performance comparison on partially labeled HSI for imbalanced classification.
3) *Configuration:* All of the verification experiments in this article were conducted in the PyTorch environment, using an NVIDIA GeForce 3090Ti 24-GB GPU. For the training process, the SGD optimizer was selected as the initial optimizer, and the learning rate was set to 0.001. The batch size was set to 64, and the number of training epochs was set to 100 for each dataset.

All methods discussed in this article (including our proposed method) are trained and evaluated on identical training, validation, and test datasets. Consistent learning rates, batch sizes, and iteration counts are applied across different frameworks to ensure experimental consistency. All experiments are conducted using identical hardware and software environments.

C. Performance Comparison

To comprehensively verify the capability of label disambiguation and class rebalance for partially labeled HSI

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TABLE I
CATEGORY INFORMATION OF THE DATASETS

Dataset		WHU-Hi-Honghu			Indian Pines			Houston 2013		
Class	Color	Class Name	Training	Total	class Name	Training	Total	class Name	Training	Total
1		Red roof	42	14041	Alfalfa	2	46	Healthy grass	63	1251
2		Road	10	3512	Corn-notill	71	1428	Stressed grass	63	1254
3		Bare soil	65	21821	Corn-mintill	41	830	Synthetic grass	35	697
4		Cotton	489	163285	Corn	12	237	Tree	62	1244
5		Cotton firewood	18	6218	Grass-pasture	24	483	Soil	62	1242
6		Rape	133	44557	Grass-trees	37	730	Water	16	325
7		Chinese cabbage	72	24103	Grass-pasture-mowed	1	28	Residential	63	1268
8		Pakchoi	12	4054	Hay-windrowed	24	478	Commercial	62	1244
9		Cabbage	32	10819	Oats	1	20	Road	63	1252
10		Tuber mustard	37	12394	Soybean-notill	49	972	Highway	61	1227
11		Brassica parachinensis	33	11015	Soybean-mintill	123	2455	Railway	62	1235
12		Brassica chinensis	26	8954	Soybean-clean	30	593	Parking Lot 1	62	1233
13		Small Brassica chinensis	57	22507	Wheat	10	205	Parking Lot 2	23	469
14		Lactuca sativa	22	7356	Woods	63	1265	Tennis Court	21	428
15		Celtuce	3	1002	Building-Grass-Trees-Drives	19	386	Runing Track	33	660
16		Film covered lettuce	21	7262	Stone-Steel-Towers	5	93			
17		Romaine lettuce	9	3010						
18		Carrot	9	3217						
19		White radish	26	8712						
20		Garlic sprout	10	3486						
21		Broad bean	3	1328						
22		Tree	12	4040						
		Total	1141	386693		512	10249		751	15029

TABLE II
CLASSIFICATION RESULTS ON THE INDIAN PINES DATASET WITH THE PARTIAL LABEL RATE 0.1

Metric	PLL(2024-2025)			Records(ICLR 2023)			HTC-LTPLL(AAAI2024)			DrPLL(ours)		
	CrCD	MAGS	CEL	SSFTT	MSSTT	MDFF	SSFTT	MSSTT	MDFF	SSFTT	MSSTT	MDFF
KC	94.65	95.75	73.50	92.57	89.75	94.92	92.05	90.60	92.41	93.80	91.22	**96.30**
OA	95.30	96.28	77.24	93.50	91.04	95.56	93.02	91.74	93.33	94.56	92.28	**96.76**
AA	96.07	87.83	55.82	75.74	69.63	84.02	80.72	76.04	80.43	83.57	81.88	**96.83**
1	100.00	54.55	56.82	0.00	0.00	15.91	0.00	0.00	0.00	0.00	97.73	**100.00**
2	95.36	93.07	72.29	94.18	92.19	93.66	89.61	87.10	89.02	93.96	91.30	94.25
3	94.93	93.79	73.64	93.41	87.58	96.32	92.14	88.59	89.35	93.16	85.93	**97.47**
4	95.58	85.33	11.50	76.11	88.50	91.59	96.90	100.00	93.81	96.46	92.04	95.58
5	92.59	96.30	86.71	79.30	89.32	98.26	77.12	91.29	93.25	79.52	94.55	**97.82**
6	94.09	99.86	93.08	100.00	95.97	97.55	98.56	97.55	96.54	100.00	96.69	98.27
7	100.00	0.00	0.00	44.44	0.00	85.19	66.67	0.00	48.15	100.00	3.70	**100.00**
8	100.00	100.00	98.46	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	**100.00**
9	100.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	89.47
10	91.56	96.42	57.79	86.26	76.08	85.82	93.83	90.69	94.81	87.01	84.52	87.12
11	95.37	99.19	94.90	97.34	97.13	99.40	93.83	91.90	93.61	96.91	92.80	98.71
12	89.89	89.70	36.35	99.29	94.68	93.09	89.01	89.54	95.74	96.63	96.63	96.28
13	99.49	99.49	78.97	98.97	97.95	97.95	97.44	96.41	100.00	98.46	98.46	97.44
14	99.00	99.83	99.08	99.58	98.67	100.00	99.75	97.25	96.92	100.00	97.59	99.92
15	98.91	97.82	33.51	91.28	83.65	94.01	96.73	97.55	96.73	96.19	89.37	98.09
16	90.36	100.00	0.00	51.69	12.36	95.51	100.00	88.76	98.88	98.88	88.76	98.88

imbalanced classification, the analysis of both quantitative experiments and visualization experiments is divided into two parts: One part is compared with the advanced PLL algorithms without the capability of class rebalance. In this portion, when focusing on the classification results of different PLL methods, the superiority of the proposed approach can be highlighted. The other part compares with the existing PLL framework with the capability of class rebalance via combining different HSI feature encoders.

1) Quantitative Results: Tables II–IV illustrate the qualitative classification results on three public and widely used HSI datasets, in which X+Y refers to the imbalanced PLL framework X with the HSI feature extraction method Y. First, taking a look into the first three columns of Tables II–IV, it iss obvious that CEL obtains the worst performance on the three datasets, because it is designed for natural image PLL and ignores the spatial–spectral information hidden in HSI. Correspondingly, due to taking the spectral–spatial characteristics into consideration, the OA of CrCD and MAGS reaches 95.30% and 96.28% respectively, which is considered relatively good, but this value is slightly lower. Nevertheless, the above three PLL methods do not account for the class imbalance in the data distribution and exhibit lower performance in terms of AA than DrPLL+MDFF. Further

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TABLE III
CLASSIFICATION RESULTS ON THE WHU-HI-HONGHU DATASET WITH THE PARTIAL LABEL RATE 0.1

Metric	PLL(2024-2025)			Records(ICLR 2023)			HTC-LTPLL(AAAI2024)			DrPLL(ours)		
	CrCD	MAGS	CEL	SSFTT	MSSTT	MDFF	SSFTT	MSSTT	MDFF	SSFTT	MSSTT	MDFF
KC	91.72	91.26	85.80	89.64	90.26	92.09	90.08	90.24	88.90	91.21	90.86	**93.55**
OA	93.43	93.11	88.93	91.82	92.33	93.77	92.15	92.27	91.19	93.07	92.78	**94.91**
AA	84.20	78.05	67.03	71.75	72.63	76.65	75.28	78.01	83.40	83.28	77.75	**85.36**
1	95.26	96.71	96.71	81.22	97.16	95.91	97.36	96.11	74.39	76.71	96.20	97.14
2	88.38	87.03	52.34	81.10	51.63	63.91	55.37	63.22	87.04	88.01	75.61	83.47
3	88.51	93.07	90.68	95.04	98.31	90.04	96.49	92.46	93.04	95.09	96.07	90.13
4	98.02	99.62	99.82	99.58	99.86	99.76	98.56	98.62	98.07	99.58	99.36	99.68
5	92.15	83.30	71.55	83.58	67.13	78.52	83.42	90.92	88.39	84.48	77.85	81.45
6	99.28	99.36	98.17	95.01	96.83	97.40	98.03	95.86	96.23	96.58	95.55	98.97
7	95.85	96.69	87.55	94.94	92.42	96.53	90.05	88.69	84.93	95.14	93.22	95.61
8	34.78	56.78	0.00	6.33	40.43	35.16	14.57	21.82	43.62	48.49	44.14	**67.37**
9	99.32	95.49	98.28	99.44	96.01	99.30	99.04	97.06	97.09	99.42	95.88	**99.68**
10	96.75	86.56	83.15	85.07	92.38	96.61	93.01	92.47	81.02	88.42	85.79	94.52
11	88.61	82.24	75.81	95.25	77.84	86.48	82.82	94.11	88.04	88.46	75.36	82.68
12	77.77	76.74	66.40	58.95	69.32	65.03	56.71	70.82	83.25	72.22	66.89	73.90
13	82.24	78.85	71.72	89.66	82.89	93.77	79.49	81.98	69.30	83.11	84.48	**90.53**
14	88.48	91.79	72.06	94.66	77.92	93.52	84.52	79.47	86.04	93.97	88.40	**94.91**
15	86.59	0.00	65.67	0.00	0.00	0.00	0.00	74.07	94.79	97.90	0.00	91.59
16	91.37	95.95	92.56	93.43	99.45	92.42	98.98	99.78	98.33	91.99	97.36	90.55
17	99.27	72.48	88.67	66.41	0.17	92.40	70.28	0.00	84.97	73.44	68.14	91.57
18	82.01	37.39	34.91	88.47	95.29	78.80	67.33	95.57	96.20	90.87	96.17	**96.29**
19	78.68	92.71	75.66	88.59	93.35	90.87	93.76	92.77	91.15	88.16	93.44	93.75
20	90.85	80.90	20.51	0.00	78.25	51.75	97.53	92.06	82.05	40.51	86.36	56.73
21	23.17	71.60	1.96	2.19	0.00	0.00	0.00	2.11	30.26	58.26	9.81	12.53
22	75.07	41.93	30.44	79.67	91.29	88.13	98.83	96.33	86.62	81.45	84.48	94.81

TABLE IV
CLASSIFICATION RESULTS ON THE HOUSTON 2013 DATASET WITH THE PARTIAL LABEL RATE 0.1

Metric	PLL(2024-2025)			Records(ICLR 2023)			HTC-LTPLL(AAAI2024)			DrPLL(ours)		
	CrCD	MAGS	CEL	SSFTT	MSSTT	MDFF	SSFTT	MSSTT	MDFF	SSFTT	MSSTT	MDFF
KC	93.55	97.55	96.35	96.28	94.42	97.77	94.92	92.02	96.36	97.13	94.47	**97.99**
OA	94.05	97.73	96.63	96.56	94.84	97.94	95.30	92.61	96.63	97.35	94.88	**98.14**
AA	93.55	97.74	96.44	96.63	94.55	98.09	95.94	93.24	96.70	97.65	94.29	**98.12**
1	86.29	97.98	99.41	100.00	91.76	99.83	75.78	77.46	91.42	100.00	93.86	98.99
2	92.11	98.66	100.00	96.81	94.63	99.58	98.32	99.50	98.49	98.49	98.66	99.33
3	96.08	100.00	100.00	100.00	100.00	100.00	100.00	99.25	100.00	100.00	100.00	**100.00**
4	90.02	99.75	96.79	98.90	94.75	98.48	98.82	92.13	99.41	98.31	95.35	99.41
5	99.75	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	**100.00**
6	86.41	98.06	87.06	90.29	85.11	100.00	99.68	88.67	99.68	100.00	85.11	98.71
7	94.69	93.78	98.17	97.68	92.28	96.35	92.70	91.54	96.51	98.26	90.62	95.19
8	84.69	94.75	86.55	85.36	94.67	96.11	90.19	90.78	92.05	85.36	89.59	**96.95**
9	95.38	95.12	90.08	84.29	92.02	94.12	92.61	95.46	99.08	90.50	88.91	93.87
10	99.40	99.91	100.00	100.00	99.66	94.12	100.00	99.49	96.40	100.00	100.00	97.17
11	95.74	97.10	99.66	100.00	94.55	100.00	99.40	96.08	97.19	100.00	96.25	**100.00**
12	97.53	99.32	93.86	99.23	92.66	100.00	98.29	73.89	94.11	99.57	97.01	**100.00**
13	95.29	91.70	95.29	97.09	86.10	93.95	93.27	95.52	86.55	95.96	78.92	92.15
14	100.00	100.00	99.75	100.00	100.00	100.00	100.00	98.77	100.00	100.00	100.00	**100.00**
15	100.00	100.00	100.00	99.84	100.00	100.00	100.00	100.00	99.68	98.25	100.00	**100.00**

observation reveals a phenomenon in keeping with expectations: the proposed DrPLL+MDFF can exceed the above three PLL methods in terms of OA, KC, and AA on three widely used partially labeled HSI scenarios. This reflects that the structure enables high-quality feature extraction through MDFF, generates more accurate category decision boundaries by the DR strategy, and achieves reliable label disambiguation via smoothing PLL loss and mutual-entropy loss. In particular, on the Indian Pines dataset, compared with CrCD and MAGS, the classification performance in terms of OA, KC, and AA improves by more than 0.48%, 0.83%, and 0.55%, respectively.

Second, when it comes to comparing class rebalance PLL frameworks, we verify the superiority of the proposed framework DrPLL under the condition of using the same feature encoder. Taking different frameworks attached to MSSTT as an example, it is obvious that the records and HTC-LTPLL attached to MSSTT only yield OA at 91.04% and 91.74% on Indiana Pines, respectively, while DrPLL with MSSTT can reach 92.28%. As can be observed, all of the OA, KC, and AA are significantly improved through DrPLL. As for the records and HTC-LTPLL attached to SSFTT, DrPLL + SSFTT still achieves better accuracy with 94.56% of OA, 93.80% of KC, and 83.57% of AA. Such improved results can be attributed to the timeliness and accuracy of the proposed dynamic class balancing strategy of DrPLL. Furthermore, it is worth noting that the improvement in terms of AA will be greater than the improvement in OA in most cases, due to the significant improvement in classification performance for the minority class.

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TABLE V

CLASSIFICATION RESULTS WITH DIFFERENT PARTIAL LABEL RATE (R) ON THE INDIAN PINES DATASET

Method	Indian Pines						
	Metrics	R=0.1		R=0.15		R=0.2	
		KC	OA	KC	OA	KC	OA
PLL	CrCD	94.65	95.30	91.87	92.81	85.99	87.55
	MAGS	95.75	96.28	94.01	94.75	93.96	94.71
	CEL	73.50	77.24	61.02	66.41	52.37	59.39
Records	SSFTT	92.57	93.50	89.30	90.66	88.92	90.31
	MSSTT	89.75	91.04	88.79	90.22	85.53	87.40
	MDFF	94.92	95.56	93.07	93.94	92.63	93.56
HTC	SSFTT	92.05	93.02	90.38	91.57	91.72	92.74
	MSSTT	90.60	91.74	89.35	90.64	89.45	90.74
	MDFF	92.41	93.33	89.63	90.91	88.58	89.98
DrPLL	SSFTT	93.80	94.56	92.31	93.27	89.91	91.16
	MSSTT	91.22	92.28	87.82	89.35	83.82	85.86
	MDFF	**96.30**	**96.76**	**95.58**	**96.13**	**94.80**	**95.44**

TABLE VI

CLASIFICATION RESULTS WITH DIFFERENT PARTIAL LABEL RATE (R) ON THE WHU-HI-HONGHU DATASET

Method	WHU-Hi-Honghu						
	Metrics	R=0.1		R=0.15		R=0.2	
		KC	OA	KC	OA	KC	OA
PLL	CrCD	91.72	93.43	87.91	90.33	86.20	88.95
	MAGS	91.26	93.11	89.70	91.86	87.11	89.92
	CEL	85.80	88.93	77.36	82.81	69.68	77.18
Records	SSFTT	89.64	91.82	87.78	90.38	86.87	89.65
	MSSTT	90.26	92.33	89.87	92.03	88.41	90.87
	MDFF	92.09	93.77	89.31	91.60	89.84	92.00
HTC	SSFTT	90.08	92.15	79.72	84.35	84.49	87.80
	MSSTT	90.24	92.27	89.70	91.86	86.93	89.64
	MDFF	88.90	91.19	86.58	89.36	84.82	87.98
DrPLL	SSFTT	91.21	93.07	87.72	90.31	85.94	88.90
	MSSTT	90.86	92.78	88.95	91.29	85.18	88.33
	MDFF	**93.55**	**94.91**	**91.53**	**93.32**	**92.02**	**93.70**

Third, focusing on the difference in classification performance of different feature encoders (e.g., MSSTT, SSFTT, MDFF) by fixing the class-rebalanced PLL framework, we discover that MDFF tends to attain better performance in most conditions. In particular, when fixing the framework as records on Indian Pines, the OA and KC of Records + MSSTT are 91.04% and 89.75%, respectively, and three categories are completely misjudged with an accuracy of 0%. The records + SSFTT gains the higher OA and KC of 93.50% and 92.57%, respectively, than records + MSSTT, yet it is approximately 2% lower than the proposed MDFF attached to records. For the WHU-Hi-Honghu and Houston 2013 datasets, the accuracy of records + MDFF is more than 1% superior to that of records + SSFTT and records + MSSTT. This phenomenon can be explained as the impact of low-quality feature representations of MSSTT and SSFTT when facing label ambiguity and class imbalance. All the imbalanced PLL frameworks connected to MDFF have the same finding in most cases, i.e., the proposed DrPLL+MDFF has the best OA, KC, and AA and does not completely misclassify any category with an accuracy of 0%.

Moreover, in order to clearly manifest the stability of the proposed method in facing varying degrees of label ambiguity, we also set the partial label rate to 0.1, 0.15, 0.2, and conduct the corresponding experiments on the Indian Pines dataset, which are listed in Tables V–VII. As shown in the tables, from the perspective of feature extraction, the proposed MDFF is more effective. From the perspective of the overall framework, the proposed DrPLL method is more adapted to the proposed MDFF in the case of increased label ambiguity, which subsequently is more effective for the imbalanced classification of partially labeled HSIs. It is obvious that the improvements on the Indian Pines and WHU-Hi-Honghu datasets are significantly larger than the Houston 2013 dataset, which is due to the fact that the former 2 have a higher degree of data imbalance than the latter.

TABLE VII

CLASIFICATION RESULTS WITH DIFFERENT PARTIAL LABEL RATE (R) ON THE HOUNSTON 2013 DATASET

Method	Houston 2013						
	Metrics	R=0.1		R=0.15		R=0.2	
		KC	OA	KC	OA	KC	OA
PLL	CrCD	93.55	94.05	90.91	91.59	91.35	92.01
	MAGS	97.55	97.73	96.84	97.07	97.21	97.42
	CEL	96.35	96.63	96.31	96.58	94.76	95.16
Records	SSFTT	96.28	96.56	95.14	95.51	95.19	95.55
	MSSTT	94.42	94.84	93.94	94.39	93.14	93.66
	MDFF	97.77	97.94	96.72	96.97	96.04	96.34
HTC	SSFTT	94.92	95.30	94.22	94.66	91.74	92.36
	MSSTT	92.02	92.61	90.93	91.61	91.75	92.36
	MDFF	96.36	96.63	93.48	93.97	92.43	93.00
DrPLL	SSFTT	97.13	97.35	95.81	96.12	94.59	94.99
	MSSTT	94.47	94.88	92.02	92.62	90.32	91.04
	MDFF	**97.99**	**98.14**	**96.53**	**96.79**	**96.75**	**96.99**

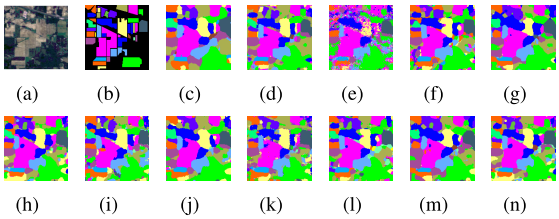


Fig. 4. Classification maps on the Indian Pines dataset. (a) Pseudocolor image. (b) Groundtruth. (c) CrCD. (d) MAGS. (e) CEL. (f) Records + SSFTT. (g) Records + MSSTT. (h) Records + MDFF. (i) HTC-LTPLL + SSFTT. (j) HTC-LTPLL + MSSTT. (k) HTC-LTPLL + MDFF. (l) DRPLL + SSFTT. (m) DRPLL + MSSTT. (n) DRPLL + MDFF.

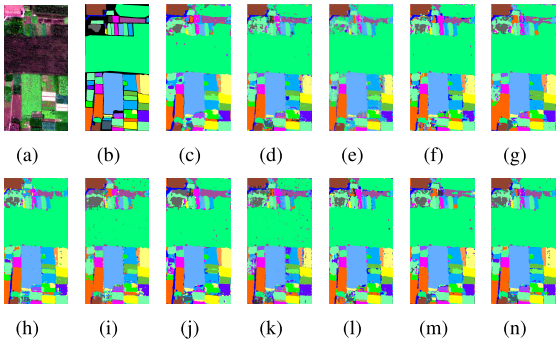


Fig. 5. Classification maps on the WHU-Hi-Honghu dataset. (a) Pseudocolor image. (b) Groundtruth. (c) CrCD. (d) MAGS. (e) CEL. (f) Records + SSFTT. (g) Records + MSSTT. (h) Records + MDFF. (i) HTC-LTPLL + SSFTT. (j) HTC-LTPLL + MSSTT. (k) HTC-LTPLL + MDFF. (l) DRPLL + SSFTT. (m) DRPLL + MSSTT. (n) DRPLL + MDFF.

2) Visualization Results: Figs. 4–6 illustrate the classification maps on the Indian Pines, WHU-Hi-Honghu, Houston 2013 datasets. First, by fixing the feature encoder, we observe the difference among different frameworks (e.g., records, HTC-LTPLL, DrPLLL). Taking the classification maps of MDFF on the Indian Pines dataset for example, records+MDFF and HTC-LTPLL+MDFF

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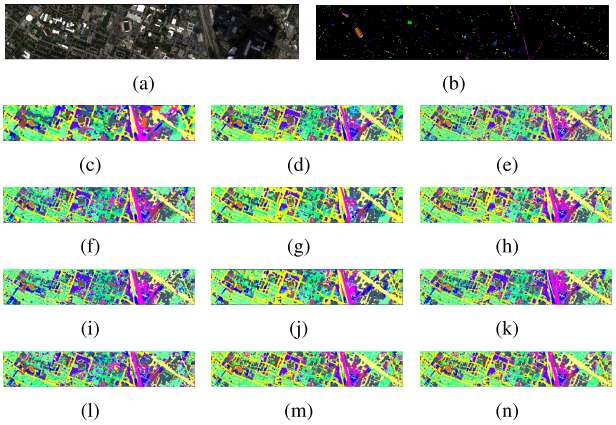


Fig. 6. Classification maps on the Houston 2013 dataset. (a) Pseudocolor image. (b) Groundtruth. (c) CrCD. (d) MAGS. (e) CEL. (f) Records + SSFTT. (g) Records + MSSTT. (h) Records + MDFF. (i) HTC-LTPLL + SSFTT. (j) HTC-LTPLL + MSSTT. (k) HTC-LTPLL + MDFF. (l) DRPLL + SSFTT. (m) DRPLL + MSSTT. (n) DRPLL + MDFF.

have varying degrees of misclassifications, resulting in visual representations that deviate from reality. By contrast, DrPLL+MDFF can achieve the clearest classification maps on the three datasets. Second, by fixing the PLL framework, it can be noted that, in most cases, MDFF harvests the more realistic classification maps than the advanced HSI feature extractor, including SSFTT and MSSTT. Taking the Indian Pines dataset as an example, records+MSSTT misclassifies minority class 7 (grass-pasture-mowed) into the majority class 5 (grass-pasture), while records+MDFF can accurately identify class 7 in most cases. It is worse that both records+SSFTT and records+MSSTT cannot identify the minority class 1 (alfalfa) precisely, which exhibits substantial discrepancies compared to the ground truth. Apparently, the mutual interaction between the proposed DrPLL and MDFF can gain the most satisfactory outcomes, which can improve the classification accuracy of the minority class while maintaining the performance of the majority class.

In conclusion, both quantitative results and visualization results from diverse perspectives demonstrate the efficacy of the proposed DrPLL framework attached to MDFF. Due to the DR strategy, the classification accuracy of the minority classes can be satisfactory. Due to the MDFF, high-quality feature representations can be acquired. In addition, the smoothing PLL loss and the mutual-entropy loss can improve the PLL capability and achieve more accurate label prediction. The proposed method in this article successfully alleviates both the label ambiguity and class imbalance, which can reduce the labeling cost and labeling pressure.

D. Parameter Analysis

In the proposed method, there are four important parameters, including the updating weight α of label confidence, the synthetic weight w of DR, and the smoothing parameter θ. In this section, we systematically analyzed the variation trend in terms of OA with different parameter settings on the Indian Pines, Houston 2013, and WHU-HI-Honghu datasets. During the parameter testing, we leveraged the control variable method, which means that only the parameters to be analyzed are varied, and the rest of the parameters are fixed.

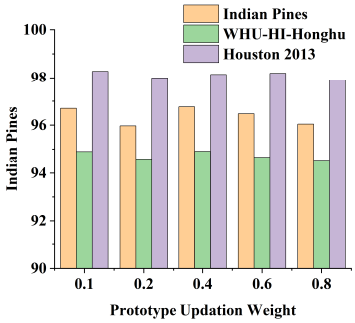


Fig. 7. OA with the variation in the updating weight α of the label prototype.

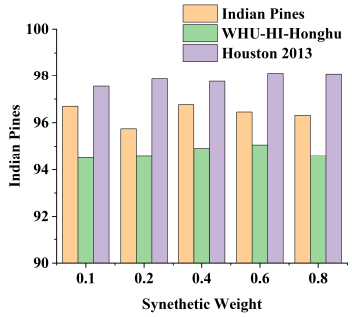


Fig. 8. OA with the variation of the synthetic weight w of DR.

1) *Updating Weight α of the Label Prototype:* α is used to dynamically update the class prototypes based on the label confidence matrix during the model training process. The impact of α on the OA of the three datasets is illustrated in Fig. 7. It is obvious that the proposed method gains the best performance when α is set to 0.5 on the Indian Pines dataset. Owing to the coarse supervision information provided by partially labeled data, the rhythm of updating label prototypes can be well controlled when α is 0.5, i.e., the class prototypes can be updated while avoiding overreliance on the current label confidence matrix, which is not exactly right during the training process.

2) *Synthetic Weight w of DR:* Fig. 8 shows the influence of the synthetic weight w of DR on the classification performance of the Indian Pines, Houston 2013, and WHU-HI-Honghu datasets. It can be observed from Fig. 8 that the relationship between the value of w and the classification performance is not a simple positive correlation. In particular, increasing the value of w within 0.1–0.4 helps to improve the classification performance and increasing the value of w within 0.6–1 is detrimental to the classification performance. This is mainly due to the fact that the value of w is too small, so the larger the proportion of category prototype I_i in the synthesized samples, the smaller the proportion of sampled samples x', resulting in the characteristics of the synthetic samples tending to be different and lacking commonalities. Conversely, if the value of w is too large, resulting in the characteristics of the synthetic samples tend to be similar and lacking diversity. In short, setting w between 0.4 and 0.6 can achieve a balance between feature commonalities and feature diversity.

3) *Smoothing Parameter θ:* The influences of the smoothing parameter θ on OA are listed in Fig. 9. From the perspective of the Indian Pines dataset, the best classification accuracy is reached with $\theta = 0.07$ and then decreases with the increasing

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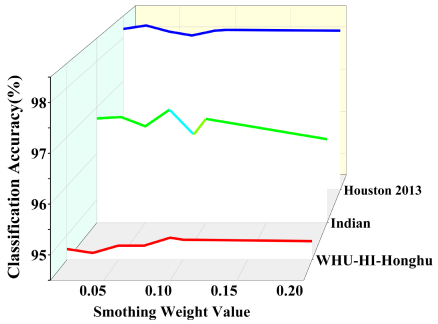


Fig. 9. OA with the variation of the smoothing parameter θ.

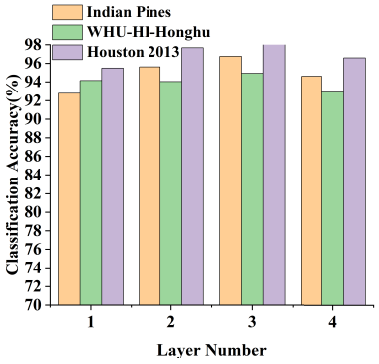


Fig. 10. OA with the variation of the layer number of multidimensional FreqFormer L.

value of θ to some extent. The model obtains the worst classification when θ is set to 0.2 on the Indian Pines dataset. Overall, proper smoothing weights are effective in preventing performance degradation due to overfitting, but too large a value of smoothing weight can also lead to model underfitting problems, so the smoothing weight on all datasets should not be set too large.

4) *Layer Number of Multidimensional FreqFormer L:* In order to further observe the impact of L on classification performance, we select L in the range of [1, 2, 3, 4] and conduct the corresponding experiments on the Indian Pines, Houston 2013, and WHU-HI-Honghu datasets. The experimental results can be seen from Fig. 10. It is obvious that the optimal performance in terms of OA is reached when $L = 3$. This is due to the fact that increasing the network depth appropriately can improve the feature extraction ability of the network model, but increasing the network depth excessively and blindly without considering the size of the data itself will bring additional noise, which in turn leads to insufficient model training and low performance.

E. Ablation Study

There are five different components, including DR, SE, MUE, SD, and MD, which means the DR, smoothing PLL loss, mutual-entropy loss, single-directional FreqFormer at channel direction, and MDFF, respectively. We set five combinations to analyze the influence of the five different components in terms of OA and KC. The whole experimental results are shown in Table VIII, where the fifth case means the

TABLE VIII

ABLATION EXPERIMENTAL RESULT ON INDINA PINES, WHU-HI-HONGHU, HOUSTON DATASETS (R = 0.2)

Case	Component					Indian Pines	
	DR	SE	MUE	SD	MD	KC	OA
1		✓	✓		✓	93.20	94.06
2	✓		✓		✓	94.37	95.06
3	✓	✓			✓	93.57	94.36
4	✓	✓	✓	✓		93.79	94.55
5	✓	✓	✓		✓	**94.80**	**95.44**
Case	Component					WHU-HI-Honghu	
	DR	SE	MUE	SD	MD	KC	OA
1		✓	✓		✓	89.26	91.56
2	✓		✓		✓	91.73	93.47
3	✓	✓			✓	91.59	93.36
4	✓	✓	✓	✓		90.78	92.72
5	✓	✓	✓		✓	**92.02**	**93.70**
Case	Component					Houston 2013	
	DR	SE	MUE	SD	MD	KC	OA
1		✓	✓		✓	96.72	96.69
2	✓		✓		✓	96.59	96.85
3	✓	✓			✓	96.42	96.69
4	✓	✓	✓	✓		95.63	95.96
5	✓	✓	✓		✓	**96.75**	**96.99**

whole framework and the other cases denote the model without corresponding components. First, the model without DR yields the worst classification results, indicating that class imbalance is one of the major bottlenecks in partially labeled HSI classification. DR alleviates the dominance of majority classes and insufficient supervision of minority classes by adaptively adjusting the sampling process according to evolving class prototypes. This reduces classification bias, improves prototype estimation and candidate label disambiguation, and provides more balanced optimization signals for all classes. Second, when SE is removed and replaced with cross-entropy loss, the classification performance drops by up to nearly 1.5% on the Indian Pines dataset. This result shows that smoothing really helps when learning with partial labels. Basically, SE stops the model from getting too confident about unclear candidate labels and helps prevent overfitting to noisy data. Softening the target distribution, it makes the model more tolerant of label ambiguity and makes optimization more stable, which leads to better overall performance. Third, when MUE is removed, both the OA and KC show a clear downward trend in the three datasets. In partially labeled HSI classification, candidate labels introduce substantial uncertainty into the prediction process. Without MUE, the model may produce inconsistent or overly entangled class probability distributions, making it difficult to distinguish the ground-truth label from confusing candidates. MUE alleviates this issue by enforcing distribution-level consistency and suppressing contradictory predictions, thus improving the reliability of pseudo-label refinement and enhancing the robustness of the disambiguation process. Finally, the SD is used instead of MD, and the OAs on the three datasets are significantly lower than the whole method in the fifth case. This illustrates that the FFT in multiple directions can mine much more useful features from HSI. HSI data exhibit highly coupled spectral–spatial structures, and discriminative information may exist along different directions of the HSI cube. A single-directional FreqFormer captures

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TABLE IX
COMPUTATION COMPLEXITY STUDY ON THE HOUSTON 2013 DATASET

Method	Params(M)	FLOPs(G)	Total Time(s)	OA
SSFTT	0.15	0.76	137.14	97.35
MSSTT	0.45	4.85	209.00	94.88
MDFF	0.53	1.3	163.04	98.14

TABLE X
PERFORMANCE COMPARISON UNDER DIFFERENT PARTIAL LABEL RATES AND MINORITY CLASS RATES ON THE HOUSTON 2013 DATASET

Partial Label Rate				
Methods	0.25		0.30	
	KC	OA	KC	OA
CrCD	87.84	88.73	86.36	87.36
MAGS	93.98	94.43	95.34	95.69
CEL	92.89	93.43	69.54	71.83
Ours	**96.45**	**96.72**	**96.06**	**96.36**
Minority Class Rate				
Methods	0.5		0.6	
	KC	OA	KC	OA
Records	96.74	96.98	96.65	96.90
HTC-HTPLL	92.72	93.26	91.13	91.77
Ours	**97.55**	**97.73**	**96.96**	**97.19**

frequency responses from only one perspective, leading to incomplete feature extraction. In contrast, MD explores the HSI cube from multiple directions in the frequency domain, enabling more comprehensive spectral–spatial representation learning for partially labeled classification. The above results demonstrate that the proposed components are complementary to each other.

F. Computation Complexity and Performance Study

As shown in Table IX, we compare the proposed feature extractor MDFF with other state-of-the-art backbone networks used in the comparison methods, including SSFTT and MSSTT, on the Houston 2013 dataset. The results demonstrate that the proposed MDFF achieves competitive classification performance while balancing the number of parameters, FLOPs, and running time, highlighting its efficient design.

G. Robustness Study

To further evaluate the robustness of the proposed method, experiments were conducted under higher partial label rates (R = 0.25 and R = 0.30) and more severe class imbalance, where the minority class rate denotes the proportion of classes containing fewer than 36 training samples. The results are reported in Table X. As the partial label rate increases, all methods suffer performance degradation due to increased label ambiguity. Nevertheless, the proposed method consistently achieves the best KC and OA, even when R = 0.30, demonstrating its strong robustness against label ambiguity. Furthermore, increasing the minority class rate results in a more severe class imbalance. Compared with existing methods, the proposed method exhibits the smallest performance degradation and consistently maintains the highest accuracy, indicating its superior capability in handling highly imbalanced data distributions.

IV. CONCLUSION

In this article, aiming at the task of HSI PLL with class imbalance, we propose a comprehensive framework, DrPLL, for partially labeled HSI imbalanced classification. Firstly, a DR strategy with class prototypes is introduced to mitigate class imbalance in DrPLL. Secondly, the MDFF is developed to efficiently extract discriminative spectral-spatial features in the frequency domain using FFT. Finally, a smoothing PLL loss and mutual-entropy loss are designed to enhance classification reliability. Extensive experiments on three benchmark datasets validate the superiority of our approach over existing methods. This approach can reduce the labeling pressure and labeling cost to some extent, but pixel-level HSI classification is still very time-consuming and labor-intensive. In future work, we will further focus on the application of easier-to-implement image-level annotation for HSI classification, which will model relationships across the entire map [42] and enable scene transitions between different scenarios [43].

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